Nucleophilic Substitution and Elimination

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1 Nucleophilic Substitution

After generating a haloalkane compound via radical halogenation, the next step is to substitute that halogen with a different atom/molecule of our choosing. One of the methods we can use to carry this out is nucleophilic substitution.

Nucleophilic substitution is a fundamental class of organic reactions where an e^- rich species, called a **nucleophile (N)**, "attacks" a molecule and replaces a specific group, known as the **leaving group (L)**. Here, the leaving group would usually be the halogen. The specific atom that is being attacked is called the **electrophile (E)**.

Nucleophilic substitution can be categorized into **bimolecular (SN2)** and **unimolecular (SN1)** nucleophilic substitution. Let's first cover SN2 reactions.

Like its name suggests, SN2 reactions are **bimolecular** reactions, meaning that both R-L and N are present in the transition state of the reaction, making the rate of the reaction determined by both [N] and [R-L]. Here is an example of a chemical equation of methyl chloride (CH_3Cl) undergoing SN2 reaction with a hydroxyl ion (OH^-).

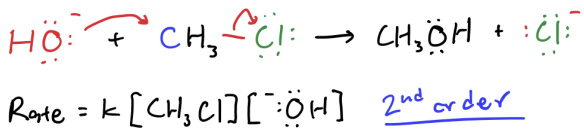

Figure 1: sn2 equation

Here, Cl is the leaving group and OH^- is the nucleophile. And when we look at the reaction from a 3D perspective, we see the following outline interaction.

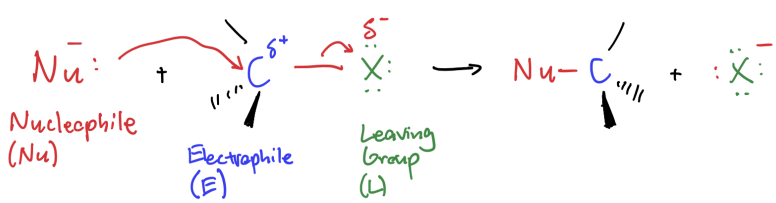

Figure 2: sn2 shell

It's important to keep in mind that SN2 reaction occurs in **one step** (bond making/breaking occurs simultaneously), meaning there are no intermediates such as those we have seen in the two step propagation of radical halogenation. As such, the energy diagram of SN2 reactions only consists of one hill and is exothermic when the Nu is a more favorable attachment over L, a topic we will discuss below. *L is denoted as X in the figure below.*

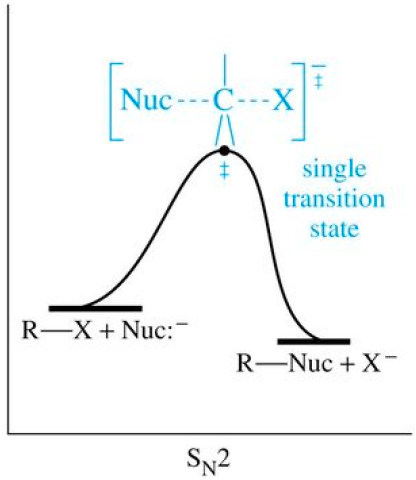

Figure 3: sn2 energy diagram

A result of reaction being one step is **stereospecificity** or, in other words, there is ALWAYS an inversion. A couple of points to keep in mind is that [1] inversion does not always imply a shift between the R and S configuration, [2] molecules show structural retention by two SN2 reactions (a.k.a. double inversion), [3] performing SN2 on only one chiral center of a molecule with two chiral centers will result is a diastereoisomer (a non-mirror-image stereoisomer), and [4] if you start with a **racemate** (50/50 mix of R/S enantiomers), SN2 will invert all of them, resulting in products that remain racemates. All of these points are intuitive with a bit of thinking about the structural consequences of the SN2 mechanism, but may not be so obvious at first glance.

Before we move onto SN1 reactions, let's take some time to define what makes a **"good" leaving group and nucleophile**.

There are two factors that determined whether a LEAVING GROUP is good: **electronegativity** and **orbital size**. In the context of EN, a good L is also a weak base A^-, which can accomodate an extra e^- well (**are stable with an extra e^-**) because they are highly electronegative. In the context of orbital size (OS), a larger OS element is more inclined to dissociate because of its relatively weak bond because their molecular bonding orbital occur at higher levels compared to elements with smaller OS. Thus better L groups appear as you go from left to right (→) and from top to bottom (↓) within the periodic table.

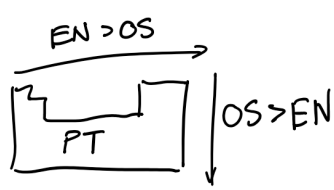

Figure 4: l ptable

Let's compare down the periodic table for halogens.

$$F^- < Cl^- < Br^- < I^-$$

The best leaving group would be I because its OS is the largest. In fact, I is such a good leaving group that it won't even participate in radical halogenation like we have seen previously. So in practice, I is not used. On the opposite end, F has the smallest OS, thus it has a much stronger bond because its bonding interactions are done through lower level orbitals, making it a much worse leaving group.

When we compare across the periodic table (row 2) we see

$$-CH_3 < -NH_2 < -OH < -F$$

The best leaving group would be F due to its greater electronegativity relative to O, N, and C. As discuessed previously, this is because the leaving group must be able to be stable with and accomadate an extra e^- when leaving the electrophile. In practice, F^- is still a poor leaving group compared to other leaving groups across the entire periodic table, HO^- is exceedingly poor, and $NH_2{}^-$ and $CH_3{}^-$ do not leave at all.

In the context of multiatomic L groups, an important factor that increases their leaving abilty is **resonance** (a.k.a. π conjugation). Recall the delocalization of e^-s in an adjacent system of p orbitals. The ability to absorb excess charge increases as the L's ability to delocalize e^-s increases. *A larger delocalization system makes the L a weaker base also.*

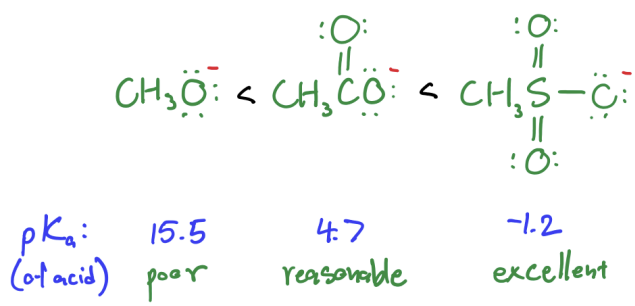

Figure 5: l delocalization

Now let's move onto NUCLEOPHILES.

Better nucleophilicity depends on four different factors: charge-basicity (for some atoms), basicity, solvation, and polaizability. Let's go through these factors one at a time.

Charge-basicity refers the favorability of an anion (compared to its cooresponding neutral state) to want to form a bond with a cation in order to delocalize its net negative charge.

$$OH^- > H_2O \qquad NH_2{}^- > NH_3 \qquad SO_4{}^{2-} > ROSO_3{}^-$$

Basicity is referring specifically to how much the nucleophile wants to share an e^- (*and thus bond*) with the electrophile. This is analogous to how it wants to share and e^- and bond with H^+. Thus elements become more nucleophilic as its electronegativity decreases (*from right to left of the periodic table*)

$$H_3N > H_2O \qquad H_2N^- > HO^- \qquad HO^- > F^-$$

Solvation refers to the ability of the surrounding solvent to impede the Nu ability to bind with the E. In polar **protic solvents** (solvents like water or ethanol that have O-H and N-H bonds) can form strong hydrogen bonds with the Nu, effectively "handcuffing" the Nu from interacting with E. Larger molecules tend to have lower charge densities because they possess a larger e^- cloud, which makes the "pull" they exert on solvent molecules much weaker and thus they are better Nu than smaller elements.

$$F^- < Cl^- < Br^- < I^-$$

In the case of polar **aprotic solvents** (solvents that are still polar but lack O-H and N-H bonds), the molecules

cannot form strong hydrogen bonds, hence the Nu is much less impeded from interacting with the E. Thus the trend completely reverses to follow basicity like we have covered previously.

$$I^- < Br^- < Cl^- < F^-$$

Mind that still in the grand scheme of things, F^- is still a weak base, it does not compare to other bases towards the left of it on the periodic table like OH^- or $NH_2{}^-$. It is simply a better base than larger elements like I^-.

Polarizability refers to the large, more diffused, and more polarizable e^- clouds of larger elements. These characteristics allows for more effective orbital overlap in the SN2 transition state compared to smaller elements. Thus

$$H_2O < H_2S < H_2Se \qquad \text{or} \qquad (CH_3)_3N < (CH_3)_3P$$

Nucleophilic ability tends to have a much more complex interplay of factors. To help digest, here is a quick summary of the points we've covered.

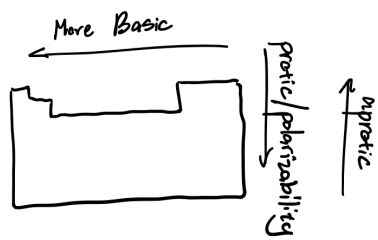

Figure 6: nuptable

At the core, nucleophilicity is driven by **basicity**, which explains the horizontal periodic trend: as you move to the left, atoms become less EN and more willing to donate their e^- density. However, the vertical trend is dictated by the "Solvent Switch." In **polar protic solvents** (like water or alcohols), smaller nucleophiles like F- are heavily "handcuffed" by a dense cage of hydrogen bonds, making larger, less-solvated ions like I- the superior attackers. Conversely, in **polar aprotic solvents** (like DMSO or acetone), these hydrogen-bonding "impediments" are absent, allowing the "naked" and inherently more basic smaller ions to regain their status as the strongest nucleophiles. Beyond basicity and solvation, **polarizability** gives larger elements a natural advantage in SN2 transition states, as their large, "squishy" electron clouds can distort and begin overlapping with the electrophilic carbon's orbital from a greater distance.

Steric effects are a factor that must be considered for both Nu and L. Steric effects refer to the e^- e^- repulsion that exist between the Nu/L and the surrounding 3 groups bonded to the target E.

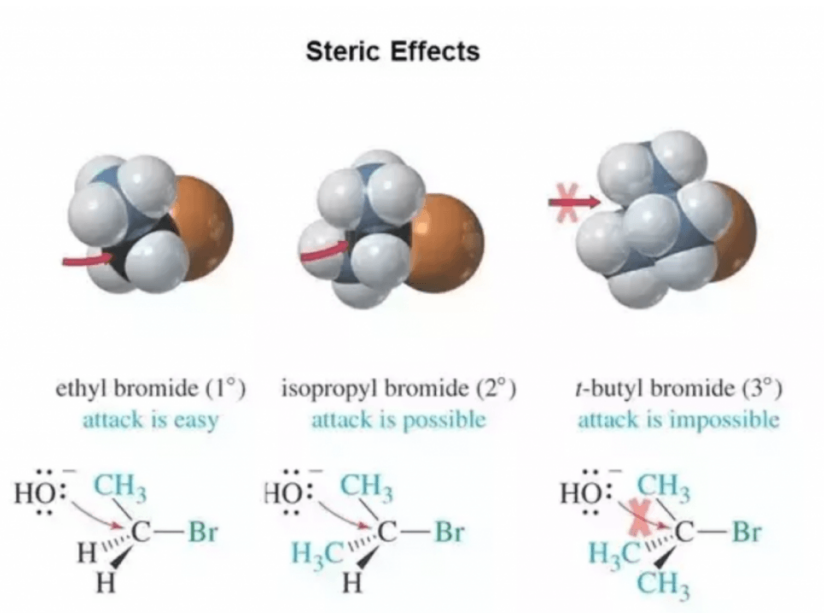

Figure 7: sterics

As a consequence of this repulsion, **leaving groups are better when they are larger** (because the repulsion is working towards pushing the L away from the E) and **nucleophiles are better when they are smaller** (because the repulsions impede its ability to attack the E as shown in the figure above).

This steric hinderance is often the greatest when the added groups are directly on the electrophile (the target carbon, α) and less impactful when added on farther neighboring atoms (β, γ, δ, etc.)

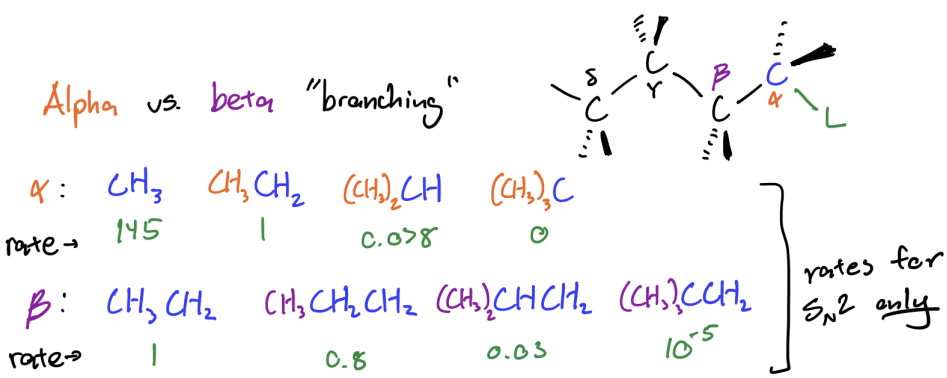

Figure 8: alpha beta branching

Okay, now let's move onto SN1 reactions. Like its name suggests, SN1 reaction are **unimolecular** reactions where the rate is solely determined by [R-L].

$$rate = k[\text{R}-\text{L}]$$

Unlike SN2, SN1 happens through a series of distinct steps.

Let us look at the outline mechanism of an SN1 reaction.

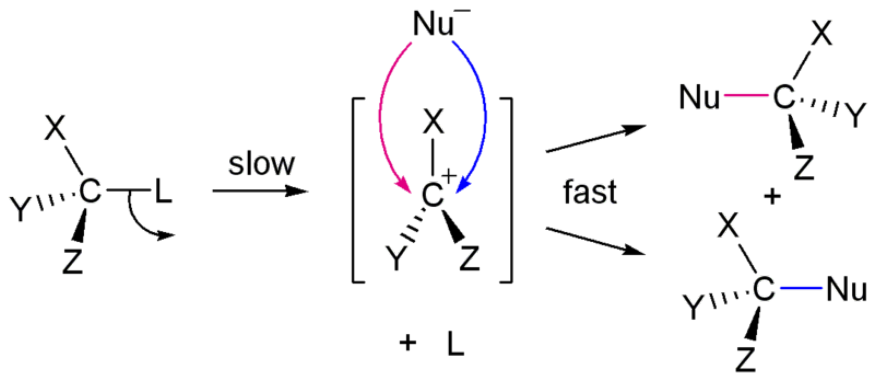

Figure 9: sn1

The very first step—the spontaneous dissociation of L from the electrophile—is the **rate limiting step** because it is highly unfavorable (although it does happen in miniscule amounts at Eq). This makes it the bottleneck of the reaction. This is followed by the nucleophilic attack of either side of the sp^2 hybridized electrophile's perpendicular p orbitals.

There are a few points that should be kept in mind regarding SN1 reactions based on its mechanism.

Firstly, as shown in the figure, the nucleophilic attack is **not stereospecific**. A pure cis (or trans) reactant pool will result in a cis/trans mix of products when put through an SN1. In practice, **slight inversion** is observed due to the lingering presence of the leaving group around the carbocation p orbital via ionic attraction, leading to a brief physical blockade for the nucleophile.

Secondly, because the reaction rate is only limited by the formation of the sp^2 hybridized carbocation, the **rate increases with a better leaving group**. Thus

$$-\mathrm{Cl} < -\mathrm{Br} < -\mathrm{I} < -\mathrm{O_3SR}$$

Thirdly, unlike SN2 reactions, **SN1 reactions accelerates with increasing polarity of solvents** (best with protic solvents). This is becasue as the leaving group (L^-) begins to depart, polar solvent molecules swarm it, forming a stable "cage" through hydrogen bonding. This "solvation" stabilizes the leaving group, making it much easier for it to leave in the first place.

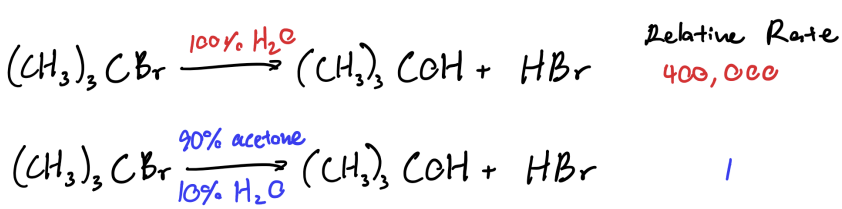

Figure 10: sn1 solvent

Notice that the previous two points are based on the same concept of improving the leaving ability of the L group to accelerate the SN1 reaction rate.

Lastly, the **product detemining steps occur after the "bottleneck"**—the formation of the carbocation. The dominant product within a solution of multiple potential nucleophiles is chosen through **competition**. The best nucleophile will bond with the intermediate to form the dominant product. For instance, take the scenario of having a mixture of NaCl and NaI in a solution of acetone.

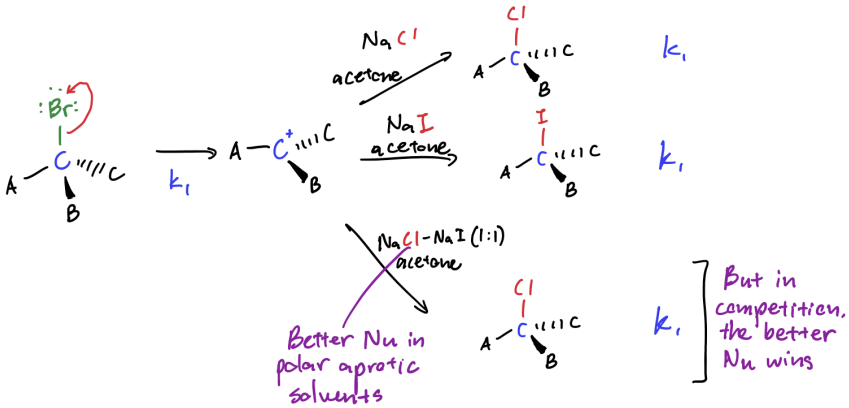

Figure 11: sn1 nu competition

Notice that the overall reaction rate is identical across all paths because it is solely dependent on the departure of the leaving group, which is constant across all three of the reactions above. And because the solvent is aprotic, Cl is a better Nu than I.

So... what makes SN1 possible? Afterall, its overall function is essentially identical to SN2. So why would a system favor SN1 over SN2 or vice versa? Two reasons.

(1) α-branching slows down competing SN2 reactions (*recall that due to steric hinderance, it is virtually impossible for a nucleophile to approach a tertiary electrophile in an SN2 manner*)

(2) α-branched carbocations are stabilized by hyperconjugation with neighboring molecular orbitals

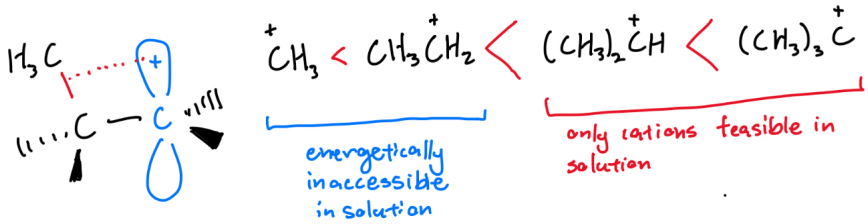

Figure 12: sn1 sn2

Here is a table nicely summarizing SN1 and SN2 reaction favorabilities across different target molecules. *Notice that the favorabilties when R is secondary depends heavily on factors like solvent and type of Nu/L rather than inherently favoring one over the other.*

R	SN1	SN2
CH_3	**Not observed** (methyl cation too high energy)	**Frequent** (fast with good Nu/L)
Primary	**Not observed** (primary carbocation too high energy)	**Frequent** (fast with good Nu/L, slow when β branching present)
Secondary	**Relatively slow** (best with good L in polar protic solvents)	**Relatively slow** (best with high concentrations of good Nu in polar aprotic solvents)
Tertiary	**Frequent** (particularly fast in polar, protic solvents and good L)	**Extremely slow**

2 Elimination

While substitution reactions were about replacing one group with another, **elimination** reactions are about **removal**. In an elimination reaction, a molecule loses two atoms or groups (typically a L group and a hydrogen atom) to form a new π bond, resulting in an alkene.

There are two types of elimination reactions: E1 and E2. Let's first look at E1.

In SN1, after the formation of a carbocation, the Nu^- can ALSO act as a base (B^-), affecting the deprotonation of a neighboring alkyl. *Mind that this scenario would not work in methane because of the absence of a neighboring alkyl.* This results in the formation of a π bonded alkene over the course of **two distinct steps** (*just like SN1*).

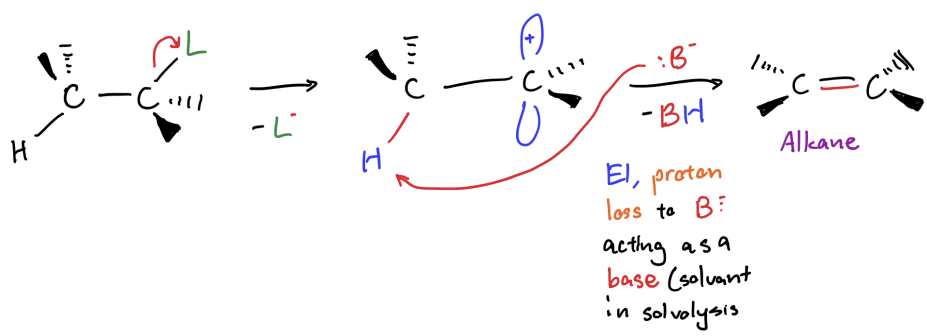

Figure 13: e1

Here the rate determining step is the same as that of SN1 and thus E1 is also a first-order reaction where its rate is only determined by [R-L]. Here is a closer look at the deprotonation step to help visualize the orbtial shifts that occur.

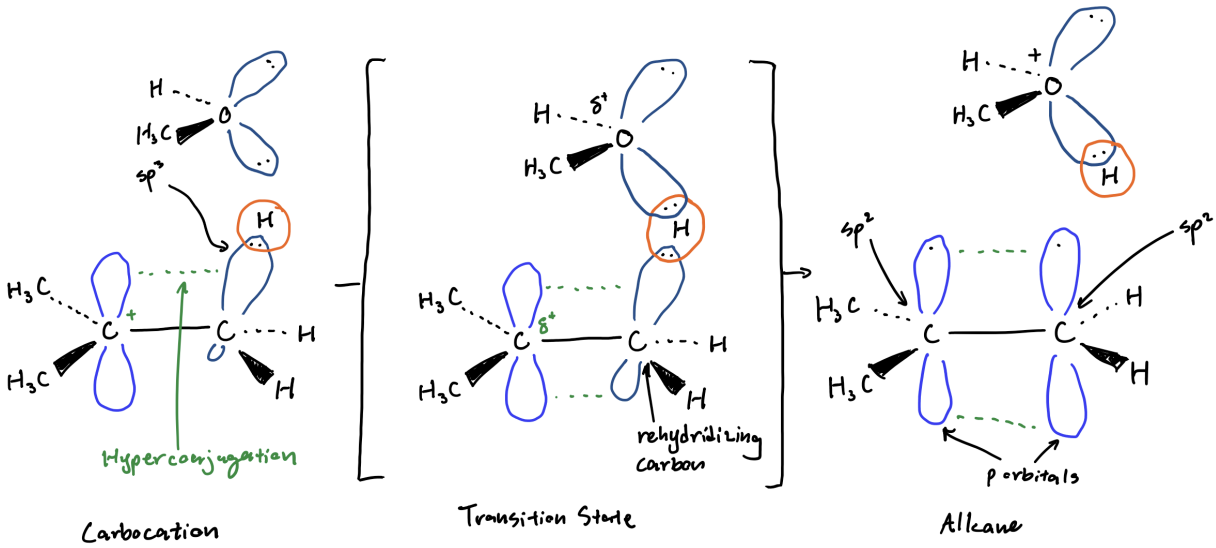

Figure 14: e1 deprotonation

And because the deprotonation step can occur to any neighboring β C-H, E1 gives us a mixture of alkanes under all of its possible scenarios (*except for a primary carbocation without a γ C*). Notice that rotation around the π bond leads to different stereoisomers.

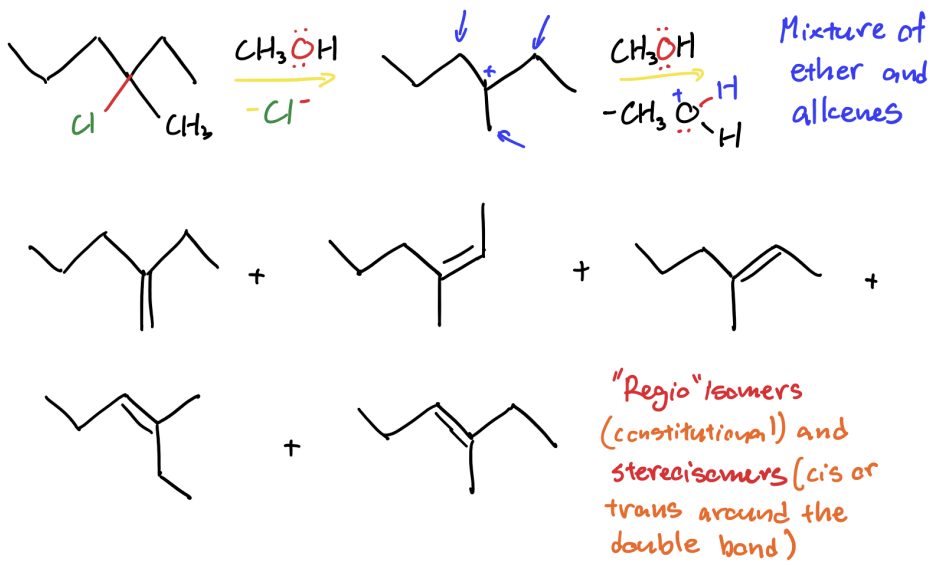

Figure 15: e1 mixture

It's important to note that both SN1 and E1 share the exact same rate-determining step: the spontaneous departure of the leaving group to form a carbocation. By the time the carbocation will undergo substitution or elimination, the L group has already departed and is out of the picture. This makes the ratio of **SN1 to E1 product independent of L**.

In that regard, there must be a way to predict a carbocation's preference between an SN1 and E1 reaction. This ratio is difficult to predict, but generally more E1 products are formed when

(1) **Higher T** because entropy of elimination is positive: HRL + B is converted to R + L + BH.

(2) The E1/SN1 ratio increases in favor of E1 when using a **poor nucleophile**. This is because SN1 requires the reagent to form a new carbon-Nu bond, necessitating higher nucleophilic 'strength.' E1 is less sensitive to this because the reagent acts as a base, simply plucking a H^+ from the molecule's exterior rather than attempting to integrate into the molecular framework.

But what about **strong bases**?

Strong bases attack the R-L **directly** at the β C-H. This is the **E2** mechanism, which is faster than SN1/E1 and also faster than SN2, if sterically hindered by α or β branching. E2 simply does each distinct step done in E1 simultaneously.

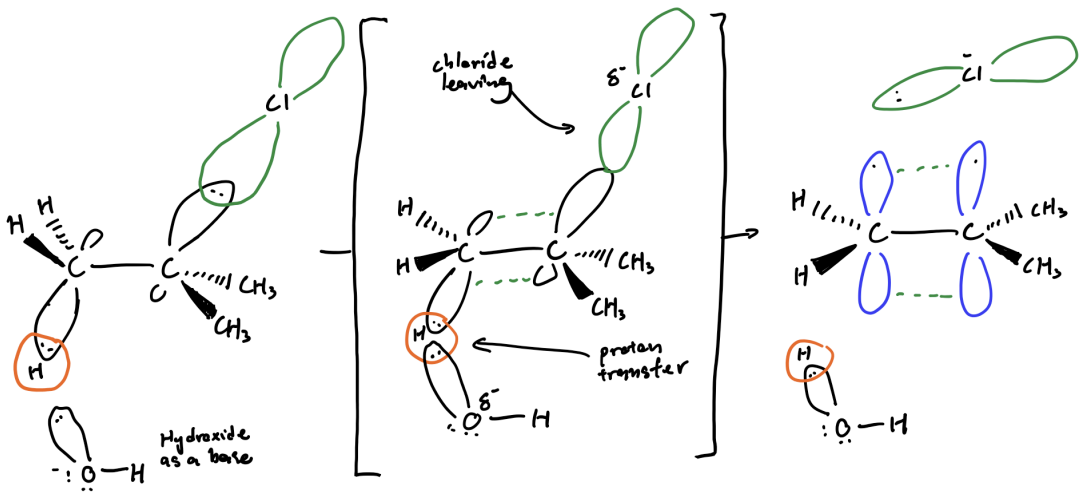

Figure 16: e2 ts

Here are a few important points about E2 reactions:

Unlike E1 reactions, E2 reactions are **second order** because the transition state depends on both the R-L and the cooresponding strong base (B).

$$Rate = k[\text{R}-\text{L}][\text{B}^-]$$

As such, a better L group and stronger B will accelerate the reaction as well.

$$\text{RCl} < \text{RBr} < \text{RI}$$

e^- withdrawing neighbors within the target molecule will accelerate E2 reactions because they make the target H more acidic by "hogging" its e^- density.

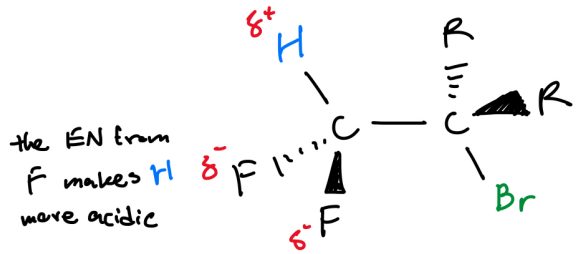

Figure 17: e2 en

Lastly, the E2 reaction is stereospecific. One diasteromer of RL will give **only one** stereoisomer of the alkane product. For instance, an RR and SS mixture will only give one product.

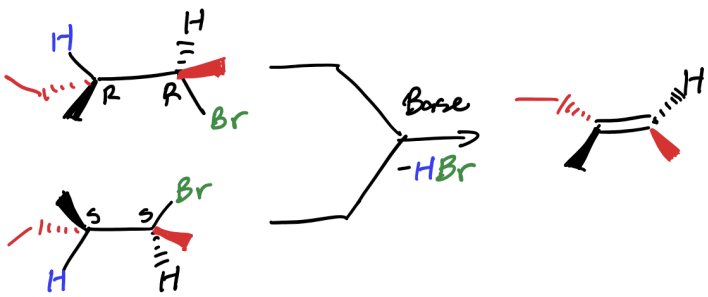

Figure 18: rrss

On the other hand, an RS and SR mixture will also give one product but a different one from the RR and SS mixture.

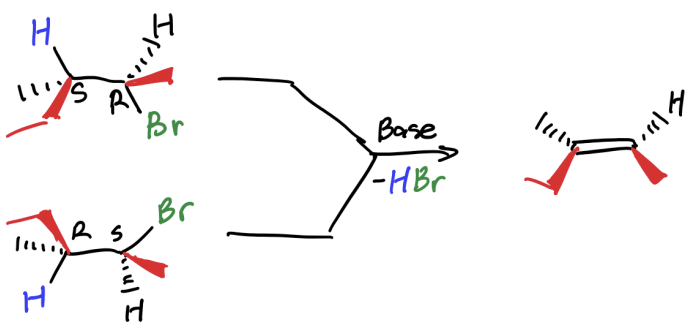

Figure 19: rssr

In E2 reactions, the H^+ and L group departure happens simultaneously to form a consequent π bond, thus the initial positions of the H^+ and L groups must be coplanar to allow the alignment of subsequent p orbitals. This can happen in one of two ways: **anti**-periplanar geometry (*a.k.a. trans*) or **syn**-periplanar geometry (*a.k.a. cis*).

In the case where the relevant C-C bond is freely able to rotate, it will preferentially be the case that it arranges itself in a anti-periplanar geometry because it minimizes steric hinderance between the L group and the H – B transitionary complex.

What about cases where the C-C bond is not freely able to rotate? such as cycloalkanes?

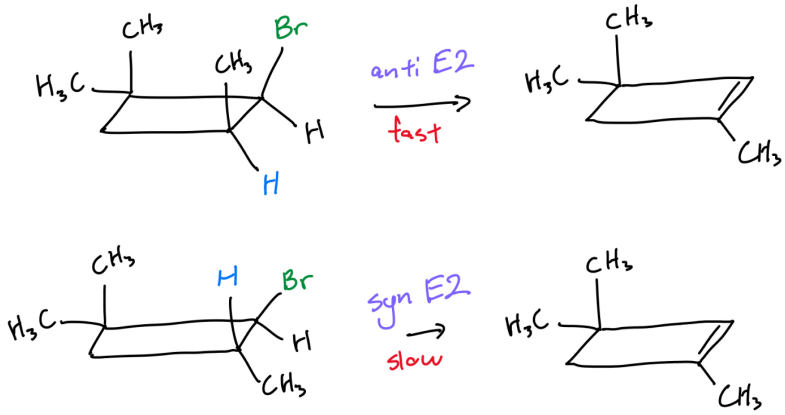

Figure 20: anti syn

The reaction proceeds but at a much slower rate due to steric hinderance between the L group and the H – B transitionary complex.

In comparison to SN2 reactions, E2 will be ensured when SN2 is slowed or shut down due to α carbon steric hinderance

3 Competition

So far, we have learned the mechanics and different characteristics of SN2, SN1, E1, and E2 reactions. Generally, they can be all be simultaneously possible across many overlapping scenarios, thus all four reactions are in competition with one another in a given reaction environment. Here is a table organizing this situation.

Type of haloalkane	**Poor Nu (e.g. H_2O)**	**Weakly basic, good Nu**	**Strongly basic, unhindered Nu**	**Strongly basic, hindered Nu**
Methyl	N/A	S_N2	S_N2	S_N2
Primary				
Unhindered	N/A	S_N2	S_N2	E2
Branched	N/A	S_N2	E2	E2
Secondary	Slow S_N1, E1	S_N2	E2	E2
Tertiary	S_N1, E1	S_N1	E2	E2

There are 3 core factors that affect the competition between SN and E.

(1) Base strength of the nucleophile

(2) Steric hinderance around the reacting carbon

(3) Steric hinderance around the nucleophile when the nucleophile is a strong base

Let's analyze across each row of the table.

PRIMARY UNHINDERED/BRANCHED: Firstly, under poor Nu circumstances, E1 nor SN1 are possible due to insufficient hyperconjugation to maintain a carbocation. SN2 nor E2 also do not work because their rate determining step is dependent on the strength of the Nu. From the "Weakly basic, good Nu" to "Strongly basic, hindered Nu", the main

determining factor is hinderance either along the haloalkane itself (via β branching) or the nucleophile (either via being a physically large or made large through solvation in protic solvents). The more hindered, E2 is preferred over SN2.

SECONDARY: From here, hyperconjugation is large enough to support the carbocation state, thus SN1 and E1 are possible under poor Nu conditions. However, E1 is more dominant because its easier for a Nu to pluck a single H^+ rather than and integrate into the haloalkane structure. For the rest of the scenarios, the haloalkane is already sterically hindered, thus the system would much rather do an E2, however it will prefer SN2 if the Nu is weakly basic enough to be uncapable of riping the H^+ off of the relevant carbon as the E2 transition state depends on this ability of the Nu/base.

TERTIARY: Due to high steric hinderance, the possibility of SN2 reactions are completely shut down, thus under weakly basic, good Nu conditions, SN1 will reign supreme. In strongly basic Nu conditions, E2 still dominates like it has done in even secondary haloalkane conditions. Under poor Nu conditions, the SN1 to E1 ratio starts to even out compared to the secondary conditions. This is because the intermediate carbocation is much more stable relative to the secondary version, thus it takes much less activation energy for the SN1 reaction to occur.

Generally speaking, rather than memorizing each of these scenarios and their cooresponding dominant mechanism, it's helpful to start with SN2 as a starting point and ask yourself the following questions.

(1) Is the Nu or haloalkane hindered? *If yes, the SN2 reaction is likely not favored.*

(2) Is the Nu strongly basic? *If yes, E2. If no, SN1/E1.*

However, keep in mind that there are other nuancial factors to consider such as absence of E1 preference for weakly basic, good Nu conditions.